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Generalized Multiscale Finite Element Method for scattering problem in heterogeneous media

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ABSTRACT

In this paper, we consider the scattering problem in the heterogeneous domain. The mathematical model is described by the Helmholtz equation related to wave propagation with absorbing boundary conditions. For the solution of the problem using the finite element method, we construct an unstructured fine grid that resolves perforation on the grid level. Such approximations lead to a large system of equations. To reduce the size of the discrete system, we use a multiscale approximation on a coarse grid using the Generalized Multiscale Finite Element Method. We construct a multiscale space using the solution of local spectral problems on the snapshot space in each local domain. Two types of multiscale basis functions are presented and investigated. We present numerical results for the Helmholtz problem in a heterogeneous domain with heterogeneous properties on obstacles. The proposed method is studied for different wavenumbers and numbers of the multiscale basis functions.

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0. Introduction

The scattering of waves by inhomogeneous highly oscillating media in the frequency domain plays an important role in physics [1]. Due to a wide range of related wave phenomena, such as dispersion and anisotropy [2,3], inhomogeneous media have found application in many fields of science and technology, including photonics [4], solid state physics [5], sound filtering [6], subwavelength imaging [7], acoustic lens design [8], cloaking [9,10] and slow light [11]. Recently, interest has appeared in the field of invisibility and cloaking, that is, suppression of electromagnetic scattering from various objects [12,13]. Invisibility cloaks have been one of the major advances in metamaterials, and several methods are now available to suppress electromagnetic scattering from various objects [14]. With its wave manipulation capabilities, metamaterials have brought invisibility one step closer to reality.

As is well known, the mesh in the standard finite element method (FEM) should: reflect the geometry and changes in material properties and be thin enough to provide the desired accuracy [15]. Thus, the computational cost of FEM is enormous for many problems, such as equations with approximate coefficients for wavenumbers and problems with highly fluctuating solutions. Therefore, it was necessary to apply such approaches to solve problems in which the classical FEM does not work or is prohibitively expensive [16]. Next, we will consider some of the main dimensionality

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reduction methods — the mathematical homogenization method, the heterogeneous multiscale method, the multiscale finite element method, and the generalized finite element method. The Mathematical Homogenization Method (MHM) has traditionally been used as the primary tool for analyzing heterogeneous environments and has been developed in previous publications [17–19]. Based on the assumptions about the periodicity of the microstructure and the homogeneity of the unit cell region, the homogenization theory divides the boundary value problem into an elementary (micro) problem and a global (macro) problem. The Heterogeneous Multiscale Method (HMM) [20–23] is seen as a general method for computing multiscale problems and has two main components, one is to choose a general macroscopic scheme such as FEM for macroscale variables at the macro-level or coarse grid, and the other in assessing the missing macro-level data by locally solving the small-scale problem. The multiscale finite element method (MsFEM) [24–26] was proposed to solve a class of elliptic problems (1) with several spatial scales arising from composite materials. Its main idea is to fix the behavior of the solution on a macroscale without resolving all microscale features by constructing multiscale functions of the shape of finite elements reflecting the local properties of the differential operator. MsFEM applies to general multiscale problems without restrictive assumptions, and the construction of shape functions is completely decoupled between the various macronutrients. The Generalized Finite Element Method (GFEM) [27,28] was developed by combining the standard FEM with the grid-based partition of unity method (PUM) [29], which allows one to construct the corresponding ansatz spaces with local properties, can effectively deal with problems with coarse coefficients or problems with highly fluctuating solutions when classical FEM approaches do not work or are too expensive. The main attributes of GFEM are the ability to use grids that are independent or partially independent of the region's geometry, and flexibility in constructing sample spaces [30].

The above considered multiscale methods, such as MHM, HMM, MsFEM and GFEM, one basis function is built for each local coarse domain to handle the effects of local inhomogeneities. However, for more complex multiscale problems, each local coarse domain contains several regions with high conductivity, and several multiscale basis functions are required to represent the local solution space. Later, a generalized multiscale finite element method (GMsFEM) was developed, which provides a framework that allows systematically enriching coarse spaces and taking into account small-scale information about inhomogeneities to construct these spaces [31–34]. The main goal of GMsFEM methods is to enrich the coarse spaces from MsFEM so that accurate coarse-scale solutions may be constructed. This method was first developed in [35]. The main idea of the GMsFEM method is to reduce computational costs by constructing multiscale basis functions on a coarse grid. The basic structure of GMsFEM contains two procedures (1) offline computing and (2) online computing. In an offline procedure, a partition of unity function is constructed, which plays a crucial role in obtaining an offline space of small dimension. Next, a set of local snapshot functions are generated to approximate the solution spaces. Snapshots are important for tasks in heterogeneous domains because they can capture the necessary information about geometry. Then local multiscale spaces are determined by performing local spectral decomposition in the image space and choosing the dominant modes in their own space. These eigenfunctions are used to form an offline space. In the online procedure, multiscale basis functions are constructed by solving local problems based on global residual information. These basic functions are calculated in such a way that they provide the greatest error reduction in offline mode. In particular, instead of using one basis per coarse node as in MsFEM, GMsFEM uses multiple multiscale basis functions per coarse node in order to more accurately capture the fine-scale effects of the system.

This paper presents a construction of the GMsFEM-based multiscale approximation for the scattering problem in the heterogeneous domain. We consider two types of multiscale basis functions based on the elliptic part of the operator and the full operator. The work is based on our previous work [24], where we presented a modified version of the MsFEM to solve the time-harmonic scattering problem. However, in MsFEM, we construct only one basis function in each local domain. In this work, we construct an enriched multiscale space based on the GMsFEM concept. Two types of multiscale basis are incorporated into a snapshot space construction, and then a local spectral problem is solved on the snapshot space to reduce the size of the local space. The proposed method is an extension of existing generalized multiscale methods. Previous approaches to developing multiscale spaces within GMsFEM for scattering problem have focused on constructing basis functions by solving a local problem based on the elliptic part of the problem operator [36]. In [37], the Edge Multiscale Methods for elliptic problems with heterogeneous coefficients is proposed. Later this method is used to design a Wavelet-based Edge Multiscale Finite Element Method for Helmholtz problems in perforated domains [38]. An Edge Multiscale Interior Penalty Discontinuous Galerkin method for heterogeneous Helmholtz problems is presented in [39]. This work is based on the regular GMsFEM and continuous approach in global coupling. We construct multiscale basis functions based on the solution of a local problem with full operator information. We observe that the proposed snapshots and partition of the unity function can accurately capture the wavenumber coefficient and consider the medium's heterogeneous properties. The paper presents numerical examples for two geometries with randomly located heterogeneities in two-dimensional domains to demonstrate the method's accuracy and effectiveness.

The work is organized as follows. In Section 1, we present the mathematical model and fine grid approximation of the problem in heterogeneous domain. In Section 2, we present a coarse grid approximation based on the Generalized Multiscale Finite Element Method. Numerical results are presented in Section 3. The paper ends with a conclusion.

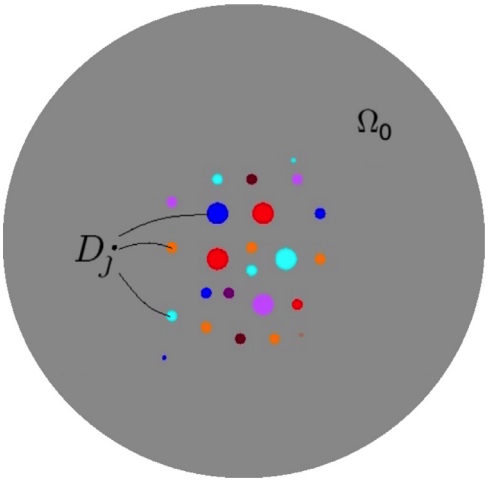


Fig. 1. Heterogeneous domain $\Omega = \Omega_0 \cup D$, $D = \bigcup_{j=1}^{M} D_j$.

1. Problem formulation

We consider the time-harmonic scattering by a heterogeneous domain in Ω

$$\Omega = \Omega_0 \cup D, \quad D = \bigcup_{j=1}^{M} D_j,$$

where D_j is the obstacles, M is the number of obstacles (see Fig. 1). We assume that outside of the scattering obstacles D_j, the material parameters are homogeneous isotropic with refractive index scaled to one.

The material properties for D_j and the homogeneous background Ω_0 is given

$$A(x) = I\big(1 - \chi_D\big) + \sum_{j=1}^{M} A_j(x)\chi_{D_j} \quad \text{and} \quad n(x) = 1\big(1 - \chi_D\big) + \sum_{j=1}^{M} n_j(x)\chi_{D_j},$$

where I is the 2-by-2 identity matrix and χ is the characteristic function ($\chi_D = \cup_{j=1}^{M}\chi_{D_j}$, $\chi_{D_j}(x)$ is equal to one if $x \in D_j$ and zero otherwise). For physical and analytical considerations we assume that $A_j(x) \in C^2\big(D_j, \mathbb{C}^{2\times 2}\big)$ is a symmetric matrix that satisfies

$$\inf_{j=1,\dots,M} \overline{\xi} \cdot \mathrm{Re}\left(A_j(x)\right)\xi \geq a_{\min}|\xi|^2 > 0 \quad \text{and} \quad \overline{\xi} \cdot \mathrm{Im}\left(A_j(x)\right)\xi \leq 0, \tag{1}$$

for almost every $x \in D$ and every $\xi \in \mathbb{C}^2$. The corresponding refractive index for the heterogeneous regions D given by $n_j(x) \in C(D_j)$ is such that

$$\inf_{j=1,\dots,M} \mathrm{Re}\big(n_j(x)\big) \geq n_{\min} > 0 \quad \text{and} \quad \mathrm{Im}\left(n_j(x)\right) \geq 0, \tag{2}$$

for almost every $x \in D_j$.

The scattering problem for the heterogeneous domain can be stated as: given the incident field

$$u^i = \mathrm{e}^{\mathrm{i}kx\cdot d} \quad \text{for} \quad |d| = 1$$

find the total field $u = u^s + u^i \in H^1_{loc}(\mathbb{R}^2)$ such that

$$\nabla A(x)\nabla u + k^2 n(x)u = 0 \quad \text{in} \quad \mathbb{R}^2, \tag{3}$$

where assume that the scattered field satisfies the Sommerfeld radiation condition at infinity. Here the wave number k is constant and assumed to be positive. Since the incident field is a solution to Helmholtz equation we can write (3) in terms of the scattered field: find the scattered field $u^s \in H^1_{loc}(\mathbb{R}^2)$ such that

$$\nabla A(x)\nabla u^s + k^2 n(x)u^s = f(x) \quad \text{in} \quad \mathbb{R}^2. \tag{4a}$$

$$\int_{\partial B_r} \left|\partial_r u^s - \mathrm{i}ku^s\right|^2 ds = o(1) \quad \text{as} \quad r \to \infty \tag{4b}$$

where B_r is the disk centered at the origin with radius $r = |x|$. Here Eq. (4b) is the known as the Sommerfeld radiation condition and the non-zero right hand side in (4a) is given by

$$f(x) = \nabla\big(I - A(x)\big)\nabla u^i + k^2\big(1 - n(x)\big)u^i$$

which is supported in the set D. It is known that (4) is well-posed do to the assumptions (1) and (2) on the coefficients (see for e.g. [40]).

Next, we formulate an appropriate boundary value problem in a truncated domain $\Omega \subset \mathbb{R}^2$. We will assume that $\overline{D} \subset \Omega$ is a bounded Lipschitz domain with boundary $\partial\Omega$ where ν is the outward unit normal vector to the boundary. To avoid having to compute the Dirichlet-to-Neumann mapping on $\partial\Omega$ we note that (4b) implies that $\partial_r u^s - \mathrm{i}ku^s = o(r^{-1/2})$ as $r \to \infty$. So we define an 'approximate' solution to (4) as: find the (approximate) scattered field $u_s \in H^1(\Omega)$ such that

$$\nabla A(x)\nabla u_s + k^2 n(x)u_s = f(x) \quad \text{in} \quad \Omega. \tag{5a}$$

$$\partial_\nu u_s - \mathrm{i}ku_s = 0 \quad \text{on} \quad \partial\Omega. \tag{5b}$$

Here in (5b) we put an absorbing boundary condition on $\partial\Omega$ to approximate the Sommerfeld radiation condition in (4b).

For approximation of (5), we use a finite element method. We have the following variational formulation for (5)

$$a(u_s, v) - \mathrm{i}b(u_s, v) - k^2 m(u_s, v) = F(v) \quad \text{for all} \quad v \in H^1(\Omega) \tag{6}$$

where the bounded sesquilinear forms on $H^1(\Omega) \times H^1(\Omega) \longmapsto \mathbb{C}$ are given by

$$a(u, v) = \int_\Omega A(x)\nabla u \cdot \nabla v \, dx, \quad m(u, v) = \int_\Omega n(x)uv \, dx, \quad b(u, v) = k\int_{\partial\Omega} uv \, ds$$

and the bounded conjugate linear functional on $H^1(\Omega) \longmapsto \mathbb{C}$ is given by

$$F(v) = \int_\Omega f v \, dx.$$

We now show that (6) is well-posed. To do so, notice that by (1) and (2) we can conclude that

$$\Big|\mathrm{Re}\,\big\{\, a(v, v) - \mathrm{i}b(v, v)\big\}\Big| \geq a_{\min}\|\nabla v\|^2_{L^2(\Omega)} \quad \text{and} \quad \Big|\mathrm{Im}\,\big\{\, a(v, v) - \mathrm{i}b(v, v)\big\}\Big| \geq k\|v\|^2_{L^2(\partial\Omega)}.$$

This gives that

$$\Big|a(v, v) - \mathrm{i}b(v, v)\Big| \geq \frac{\min\{k, a_{\min}\}}{2}\Big\{\|\nabla v\|^2_{L^2(\Omega)} + \|v\|^2_{L^2(\partial\Omega)}\Big\}$$

for all $v \in H^1(\Omega)$. Notice that it can be shown that the mapping

$$\varphi \longmapsto \|\nabla\varphi\|^2_{L^2(\Omega)} + \|\varphi\|^2_{L^2(\partial\Omega)}$$

is equivalent to the standard $H^1(\Omega)$ norm by applying a standard Poincarè type contradiction argument. Therefore, we have that the sesquilinear form $a(\cdot\,,\cdot) - \mathrm{i}b(\cdot\,,\cdot)$ is coercive on $H^1(\Omega)$. By appealing to the compact embedding of $H^1(\Omega)$ into $L^2(\Omega)$ we can conclude that the sesquilinear form $m(\cdot\,,\cdot)$ is compact. This implies that (6) is the compact perturbation of an invertible operator which implies that is (6) satisfies the Fredholm alternative.

Theorem 1. *For any given incident field u^i with $|d| = 1$ there exists a unique solution $u_s \in H^1(\Omega)$ to (5) and equivalently (6) such that $\|u_s\|_{H^1(\Omega)} \leq C\|u^i\|_{C^2(\Omega)}$ where the constant $C > 0$ is independent of the scattered and incidents fields.*

Proof. Notice that we have already shown that $a(\cdot\,,\cdot) - \mathrm{i}b(\cdot\,,\cdot) - k^2 m(\cdot\,,\cdot)$ is a compact perturbation of an invertible operator which implies that injectivity implies solvability. We now prove that (6) has a unique solution. To this end, assume $w \in H^1(\Omega)$ satisfies

$$a(w, v) - \mathrm{i}b(w, v) - k^2 m(w, v) = 0 \quad \text{for all} \quad v \in H^1(\Omega).$$

By letting $v = w$ and taking the imaginary part of above equality gives that

$$0 = -\mathrm{Im}\,\big\{\, a(w, w) - \mathrm{i}b(w, w) - k^2 m(w, w)\big\} \geq k\|w\|^2_{L^2(\partial\Omega)}$$

where we have used (1) and (2). We can conclude that $w = 0$ on $\partial\Omega$ and integration by parts implies that w satisfies (5a) with $f = 0$. Now (5b) gives that w has zero Cauchy data on $\partial\Omega$ and uniqueness follows from the unique continuation principal. This proves that existence and uniqueness of the solution for any F is the dual space of $H^1(\Omega)$. We can conclude that the sesquilinear form $a(\cdot\,,\cdot) - \mathrm{i}b(\cdot\,,\cdot) - k^2 m(\cdot\,,\cdot)$ satisfies the inf–sup condition. Therefore, we have that there is a constant $\alpha > 0$ such that the solution u_s to (6) satisfies

$$\alpha\|u_s\|_{H^1(\Omega)} \leq \sup_{\|v\|_{H^1(\Omega)}=1} \Big|a(u_s, v) - \mathrm{i}b(u_s, v) - k^2 m(u_s, v)\Big| = \sup_{\|v\|_{H^1(\Omega)}=1} |F(v)|.$$

and the a prior estimate is given by the fact that one can easily show that $|F(v)| \leq C\|u^i\|_{C^2(\Omega)}\|v\|_{H^1(\Omega)}$ for all $v \in H^1(\Omega)$. Proving the claim. □

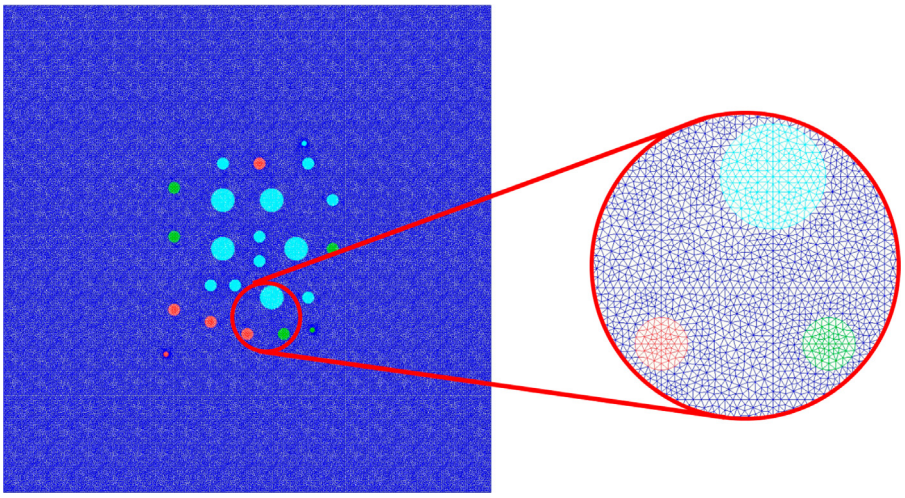


Fig. 2. Illustration of the fine grid $\mathcal{T}_h$ for heterogeneous domain Ω.

Let $\mathcal{T}_h$ be an unstructured fine grid that resolve small scale heterogeneities (obstacles) on grid level (see Fig. 2). Notice that as a consequence of the above result we have that for any standard finite element $V_h \subset H^1(\Omega)$ (where h =mesh size) the variational problem: find $u_s^h \in V_h$ such that

$$a\big(u_s^h, v\big) - \mathrm{i}b\big(u_s^h, v\big) - k^2\, m\big(u_s^h, v\big) = F(v) \quad \text{for all} \quad v \in V_h \tag{7}$$

is well-posed since the sesquilinear form $a(\cdot\,,\cdot) - \mathrm{i}b(\cdot\,,\cdot) - k^2 m(\cdot\,,\cdot)$ satisfies the inf–sup condition. Therefore, we have that for any V_h that satisfies

$$\inf_{w_h \in V_h} \|w - w_h\|_{H^1(\Omega)} \longrightarrow 0 \quad \text{as} \quad h \to 0^+ \quad \text{for all} \quad w \in H^1(\Omega)$$

then we can conclude that

$$\|u_s - u_s^h\|_{H^1(\Omega)} \longrightarrow 0 \quad \text{as} \quad h \to 0^+$$

where u_s is the solution to (6) and u_s^h is the solution to (7) since the inf–sup condition and approximation property of V_h implies convergence.

Letting V_h be the standard piece-wise linear finite element space gives that

$$u_s^h = \sum_j U_j \phi_j,$$

where ϕ_j are the standard linear basis functions.

We can write (7) in the matrix form

$$\big(K_h - \mathrm{i}\, B_h - k^2 M_h\big) U_h = F_h, \tag{8}$$

where $[M_h]_{i,j} = m\big(\phi_i, \phi_j\big)$ is the mass matrix, $[B_h]_{i,j} = b\big(\phi_i, \phi_j\big)$ is the boundary mass matrix and $[K_h]_{i,j} = a\big(\phi_i, \phi_j\big)$ is the stiffness matrix for $i, j = 1, \ldots, \dim(V_h)$. Here the forcing vector $[F_h]_j = F(\phi_j)$ and the solution vector $[U_h]_j = U_j$.

2. Multiscale approximation using GMsFEM

For coarse-scale approximation, we use Generalized Multiscale Finite Element Method. In the GMsFEM, we have following computational algorithm:

- the construction of the multiscale basis functions by the solution of the local eigenvalue problem in local domain ω_p and
- the construction and solution of the coarse grid approximation on multiscale space.

We define coarse grid $\mathcal{T}_H$ in domain Ω with mesh sizes $H \gg h > 0$

$$\mathcal{T}_H = \cup_{p=1}^{N_c} K_p,$$

where K_p is local domain, N_c is number of coarse cells (see Fig. 3). We note that the fine-scale triangulation $\mathcal{T}_h$ is conforming with coarse-grid edges. Let V_H be a finite-dimensional function space, which consists of functions that are

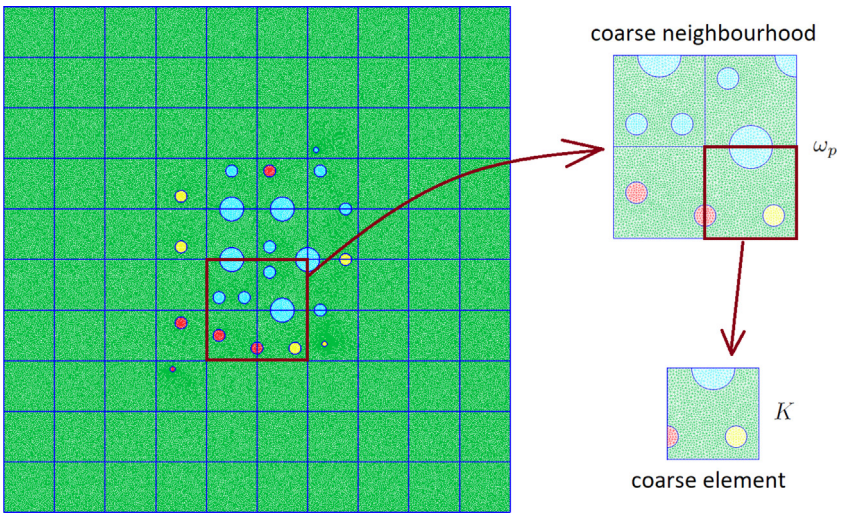


Fig. 3. Illustration of coarse grids and local domain ω_p.

smooth on each local domain related to the coarse mesh vertices. We construct local multiscale space V_H by solution of the local eigenvalue problems on each local domain ω_p, $p = 1, \dots, N_v$, N_v is number of coarse vertices. Furthermore, eigenfunctions corresponding to the dominant eigenvalues are used as the multiscale basis functions.

The multiscale space V_H is then defined as the linear span of all $V_H^{\omega_p}$, $\omega_p \in \mathcal{T}_H$ and will be used as the approximation space of continuous Galerkin approach, which can be formulated as follows: find $u_H \in V_H$:

$$a(u_H, v) - ib(u_H, v) - k^2 m(u_H, v) = F(v), \quad v \in V_H$$

Next, we define GMsFEM algorithm in details. In this project we focus on the two-dimensional formulation and use a structured coarse grid with quadratic coarse cells. Therefore, the local domain ω_p is defined as a union of coarse cells around coarse-grid vertex. In multiscale method, we have following steps (see Fig. 4 for illustration):

1. Generation of the fine grid that conforming with the coarse grid edges and extraction of the submesh for local domain ω_p.
2. Solution of the local problems in each local domain ω_p with different boundary conditions (snapshot space) and dimension reduction by solution of the local spectral problem on the snapshot space ($\Psi_\kappa^{\omega_p}$).
3. Generation of the local multiscale basis functions by multiplication of the dominant eigenvalues $\Psi_\kappa^{\omega_p}$ from previous step ($p = 1, \dots, M_i$) to the multiscale partition of unity function χ_p ($\phi_\kappa^{\omega_p} = \chi_p \Psi_\kappa^{\omega_p}$).
4. Construction of the projection matrix $R = (R_1, R_2, \dots, R_{N_v})^T$, $R_i = (\phi_1^{\omega_p}, \phi_2^{\omega_p}, \dots, \phi_{M_i}^{\omega_p})^T$.
5. Generation of the coarse scale matrices and right-hand side, $M_H = RM_hR^T$, $K_H = RK_hR^T$, $B_H = RB_hR^T$, $F_H = RF_h$ using projection matrix, fine-scale matrices and right-hand side from (8).
6. Solution of the coarse-scale system, $(K_H - iB_H - k^2 M_H)U_H = F_H$.
7. Reconstruction of the fine-scale solution, $U_{ms} = R^T U_H$.

We start with the definition of the snapshot space in ω_p and present construction of the multiscale basis functions in details. We present construction of the two types of multiscale basis functions. First type of basis functions is based on the elliptic part of problem operator ($\mathcal{L}u = \nabla A(x)\nabla u$) for snapshot space and multiscale partition of unity functions construction. Second type is based on the full operator ($\mathcal{L}u = \nabla A(x)\nabla u + k^2 n(x)u$). We note that, the spectral problem that used for determination of the dominant modes is similar for both types.

Snapshot space. To construct the local snapshot space $V^{\omega_p,\text{snap}}$, we find $\psi_l^{\omega_p} \in V_h(\omega_p)$ by solving the following local problem in local domain ω_p

- *Type 1.*

$$\begin{aligned} \nabla A(x)\nabla \psi_l^{\omega_p} &= 0 \quad \text{in} \quad \omega_p, \\ \psi_l^{\omega_p} &= \delta_l \quad \text{on} \quad \partial\omega_p. \end{aligned} \tag{9}$$

- *Type 2.*

$$\begin{aligned} \nabla A(x)\nabla \psi_l^{\omega_p} + k^2 n(x)\psi_l^{\omega_p} &= 0 \quad \text{in} \quad \omega_p, \\ \psi_l^{\omega_p} &= \delta_l \quad \text{on} \quad \partial\omega_p. \end{aligned} \tag{10}$$

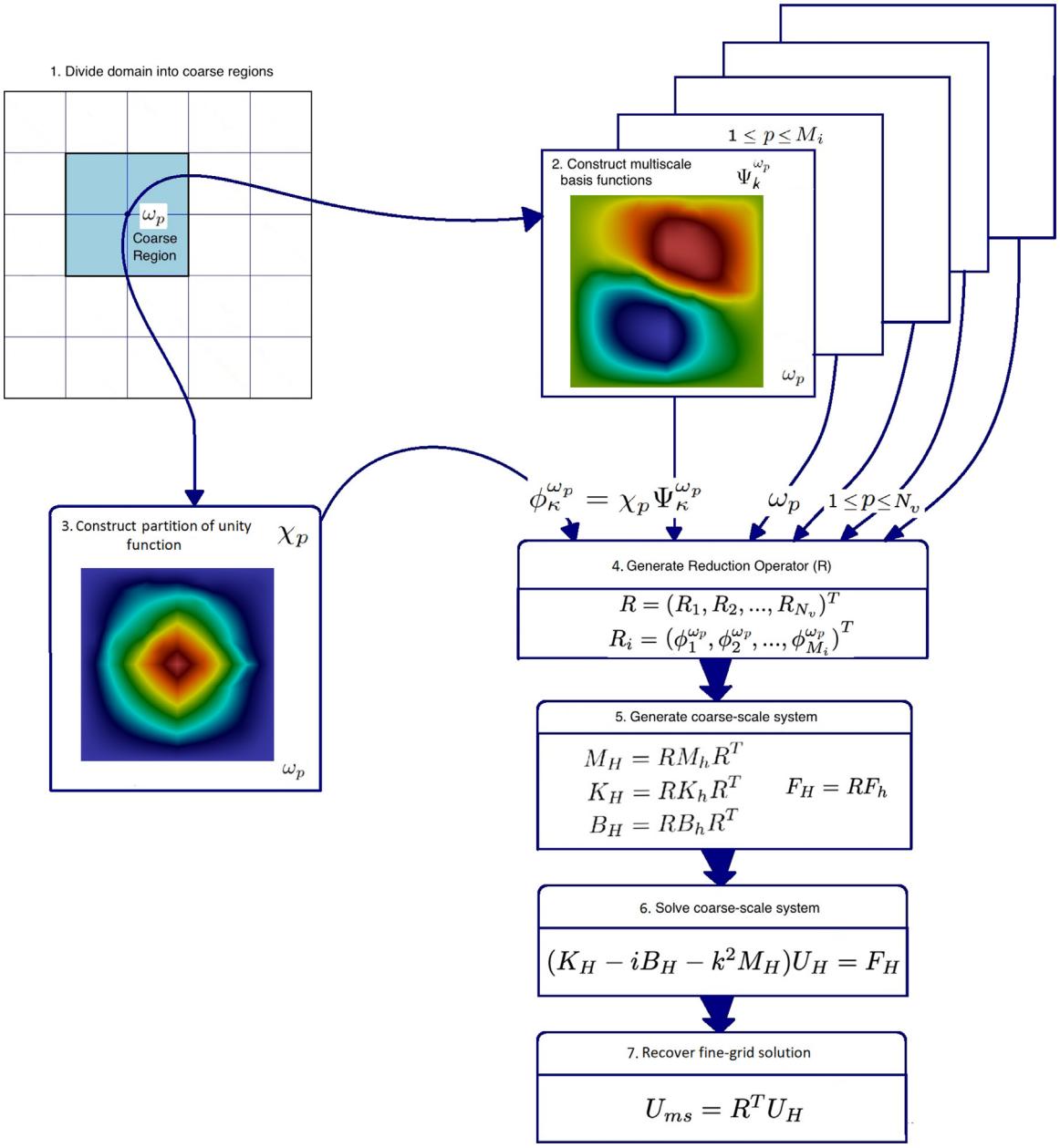


Fig. 4. Illustration of the GMsFEM algorithm.

Here δ_l is the delta function that is equal to 1 on one boundary fine-grid vertex and 0 otherwise ($l = 1, \ldots, J_i$ with J_i being the total number of vertices on boundary). The functions $\psi_l^{\omega_p}$ defined above are the harmonic extensions. The linear span of the harmonic extensions is the local snapshot space

$$V^{\omega_p,\text{snap}} = \text{span}\{\psi_1^{\omega_p}, \ldots, \psi_{J_i}^{\omega_p}\}. \tag{11}$$

Multiscale space. We select a few important modes from $V^{\omega_p,\text{snap}}$ to form a local multiscale space $V_H(\omega_p)$. The important modes are obtained from the following local spectral problem defined in the local snapshot space

$$A^{\omega_p,\text{snap}}\Psi^{\omega_p,\text{snap}} = \lambda\, S^{\omega_p,\text{snap}}\Psi^{\omega_p,\text{snap}}, \tag{12}$$

where

$$A^{\omega_p,\text{snap}} = R^{\omega_p,\text{snap}}A^{\omega_p}(R^{\omega_p,\text{snap}})^T, \quad S^{\omega_p,\text{snap}} = R^{\omega_p,\text{snap}}S^{\omega_p}(R^{\omega_p,\text{snap}})^T,$$

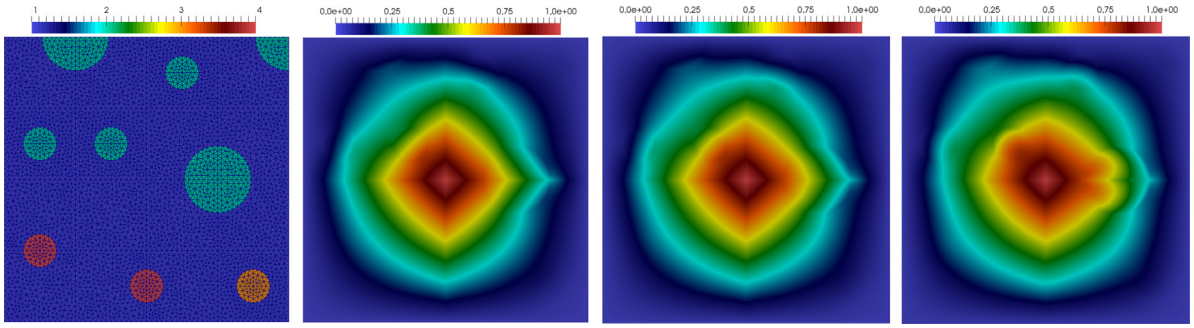


Fig. 5. Illustration of the local domain and partition of unity functions. First: local domain ω_p, $p = 49$. Second: *Type 1* multiscale partition of unity function. Third: *Type 2* multiscale partition of unity function for $k = 1$. Fourth: *Type 1* multiscale partition of unity function for $k = 2$.

$$A^{\omega_p} = [a_{mn}] = a(\phi_m, \phi_n) = \int_{\omega_p} A(x)\nabla\phi_m \cdot \nabla\phi_n dx,$$

$$S^{\omega_p} = [s_{mn}] = s(\phi_m, \phi_n) = \int_{\omega_p} A(x)\phi_m\phi_n dx,$$

and projection matrix into the snapshot space is defined as follows

$$R^{\omega_p,\text{snap}} = (\psi_1^{\omega_p}, \dots, \psi_{J_i}^{\omega_p})^T.$$

To construct a reduced space $V_H(\omega_p)$, we select the first M_i eigenvectors $\Psi_1^{\omega_p,\text{snap}}, \Psi_2^{\omega_p,\text{snap}}, \dots, \Psi_{M_i}^{\omega_p,\text{snap}}$ corresponding to the first M_i smallest eigenvalues $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_{M_i}$, and define the space $V_H(\omega_p)$ by

$$V_H(\omega_p) = \text{span}\{\phi_1^{\omega_p}, \dots, \phi_{M_i}^{\omega_p}\}. \tag{13}$$

where $\phi_\kappa^{\omega_p} = \chi_p \Psi_\kappa^{\omega_p}$ and $\Psi_\kappa^{\omega_p} = (R^{\omega_p,\text{snap}})^T \Psi_\kappa^{\omega_p,\text{snap}}$, $k = 1, \dots, M_i$ (see Fig. 6). Here χ_p is the multiscale partition of unity functions.

Multiscale partition of unity functions. To construct the partition of unity functions, we solve following local problem in $\forall K_j \in \omega_p$ (see Fig. 5). :

- *Type 1*.

$$\begin{aligned} \nabla A(x)\nabla\chi_p &= 0 \quad \text{in} \quad K_j, \\ \chi_p &= g_p \quad \text{on} \quad \partial K_j. \end{aligned} \tag{14}$$

- *Type 2*.

$$\begin{aligned} \nabla A(x)\nabla\chi_p + k^2 n(x)\chi_p &= 0 \quad \text{in} \quad K_j, \\ \chi_p &= g_p \quad \text{on} \quad \partial K_j. \end{aligned} \tag{15}$$

Here g_p is the linear basis functions defined on the coarse grid.

Global projection approach for definition of the coarse-scale system. The coarse-scale system can be calculated by projecting the fine-scale matrices onto the coarse grid with the global projection matrix assembled from the calculated multiscale basis functions

$$R = (R_1, R_2, \dots, R_{N_v})^T, \quad R_p = (\phi_1^{\omega_p}, \phi_2^{\omega_p}, \dots, \phi_{M_i}^{\omega_p})^T \tag{16}$$

where R_p is the local projection matrix in a local domain ω_p and N_v is the number of coarse grid vertices.

In the numerical implementation, we first assemble the global matrices M_h, K_h, B_h and vector F_h in (8). Then using the global projection matrix R, we define the coarse-scale system

$$(K_H - iB_H - k^2 M_H)U_H = F_H, \tag{17}$$

where $M_H = RM_hR^T$ is the coarse-scale mass matrix, $K_H = RK_hR^T$ is the coarse-scale stiffness matrix, $B_H = RB_hR^T$ is the coarse-scale boundary mass matrix and $F_H = RF_h$. is the coarse-scale right-hand side. After calculation of the coarse-scale solution U_H, we can recover the fine-scale solution by $U_{ms} = R^T U_H$.

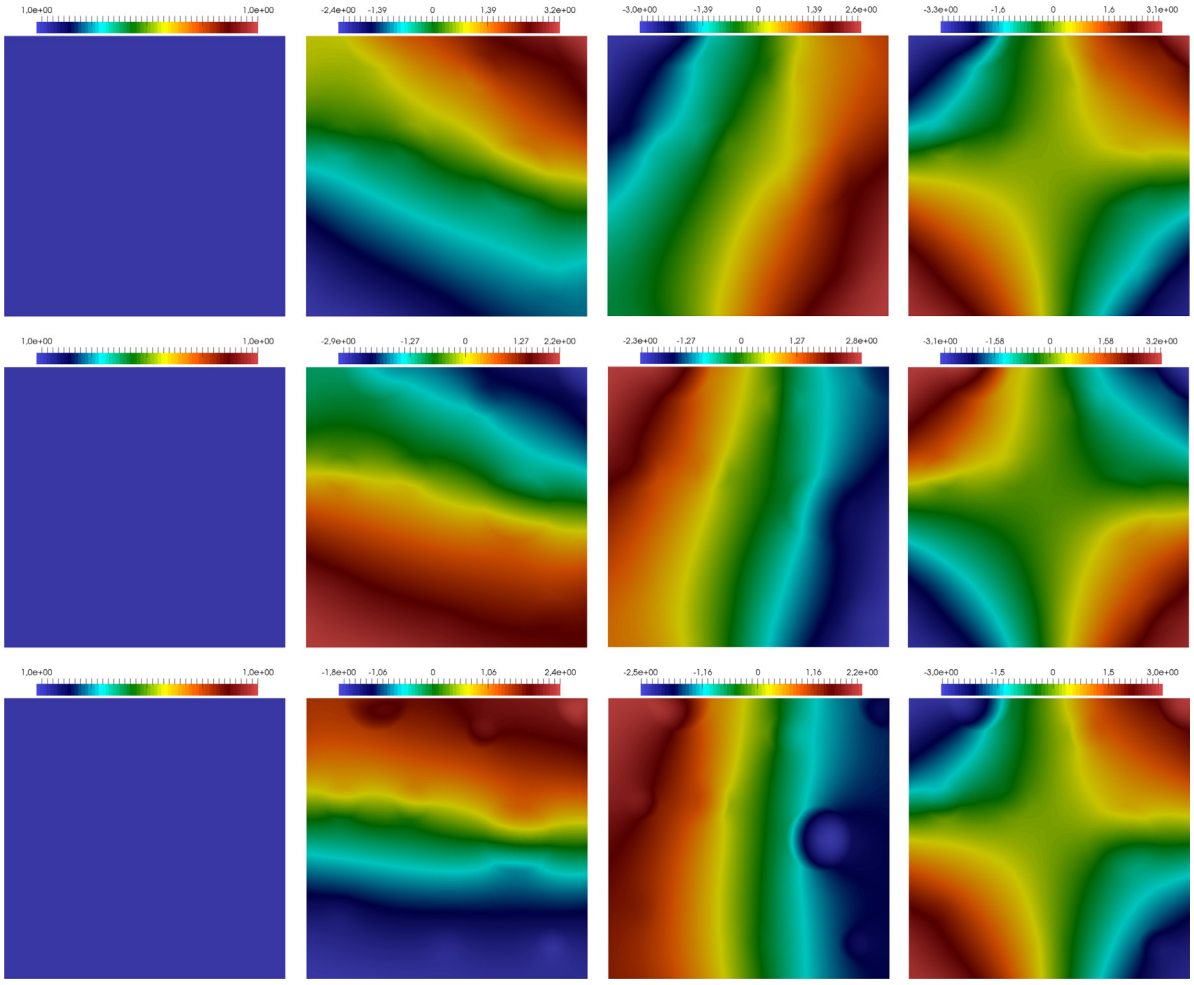


Fig. 6. Illustration of eigenvectors corresponded to the first four smallest eigenvalues in local domain ω_p, $p = 49$. First: *Type 1* spectral problem. Second: *Type 2* spectral problem with $k = 1$. Third: *Type 1* spectral problem with $k = 2$.

3. Numerical results

We present results for fine-scale and coarse-scale solutions using presented multiscale method. For numerical investigation, we use a fine-grid solution as a reference solution for errors calculation. We consider two examples of the location and size of heterogeneities to verify the effectiveness of GMsFEM method for the numerical simulation of a coarse grid.

We consider problem (5) in domain $\Omega = [-2, -2] \times [2, 2]$ with several heterogeneous obstacles. We set $k = 1.0$ and $d = (1, 0)$. For background media, we set $\alpha = 1$ and $n = 1$ in Ω_0.

To investigate the proposed multiscale method, we consider two structured coarse grids

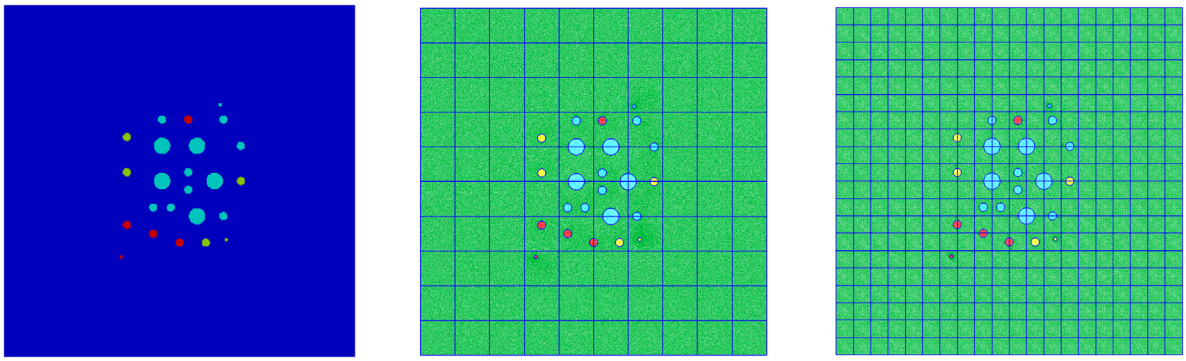
(1) 10×10 coarse grid with 121 vertices and 100 quadratic elements,
(2) 20×20 coarse grid with 441 vertices and 400 quadratic elements.

Note that, the fine grids are conforming with inclusions and coarse grid edges. The computational implementation is based on the FEniCS library [41]. The GMSH program [42] is used to construct a geometric area and corresponded computational grids. The results are visualized using the Paraview program [43].

To compare the results, we calculate relative errors in L^2 norm between multiscale solution and fine-scale solution

$$e(u) = \frac{\|u - u_f\|_{L_2}}{\|u\|_{L_2}} \cdot 100\%, \quad \|u\|_{L_2} = \sqrt{\int_\Omega u^2 \, dx}$$

where $e_{Re} = e(Re(u))$, $e_{Im} = e(Im(u))$, $e_{Abs} = e(Abs(u))$, u is multiscale solution using GMsFEM and u_f is reference (fine grid) solution.



(a) Computational domain. (b) 10×10 coarse mesh. (c) 20×20 coarse mesh.

Fig. 7. Computational domain, fine and coarse meshes. (a) Computational domain with obstacles $\Omega = \Omega_0 \cup D$. Ω_0 is the backround media (blue color). D is the obstacles (blue color — first type, green — second type and red — third type). Fine mesh with 100 802 vertices and 200 482 triangle elements (green color for background media and multicolor for obstacles). (b) 10×10 coarse mesh with 121 vertices and 100 quadratic elements (blue color). (c) 20×20 coarse mesh 441 vertices and 400 quadratic elements (blue color). Example 1.

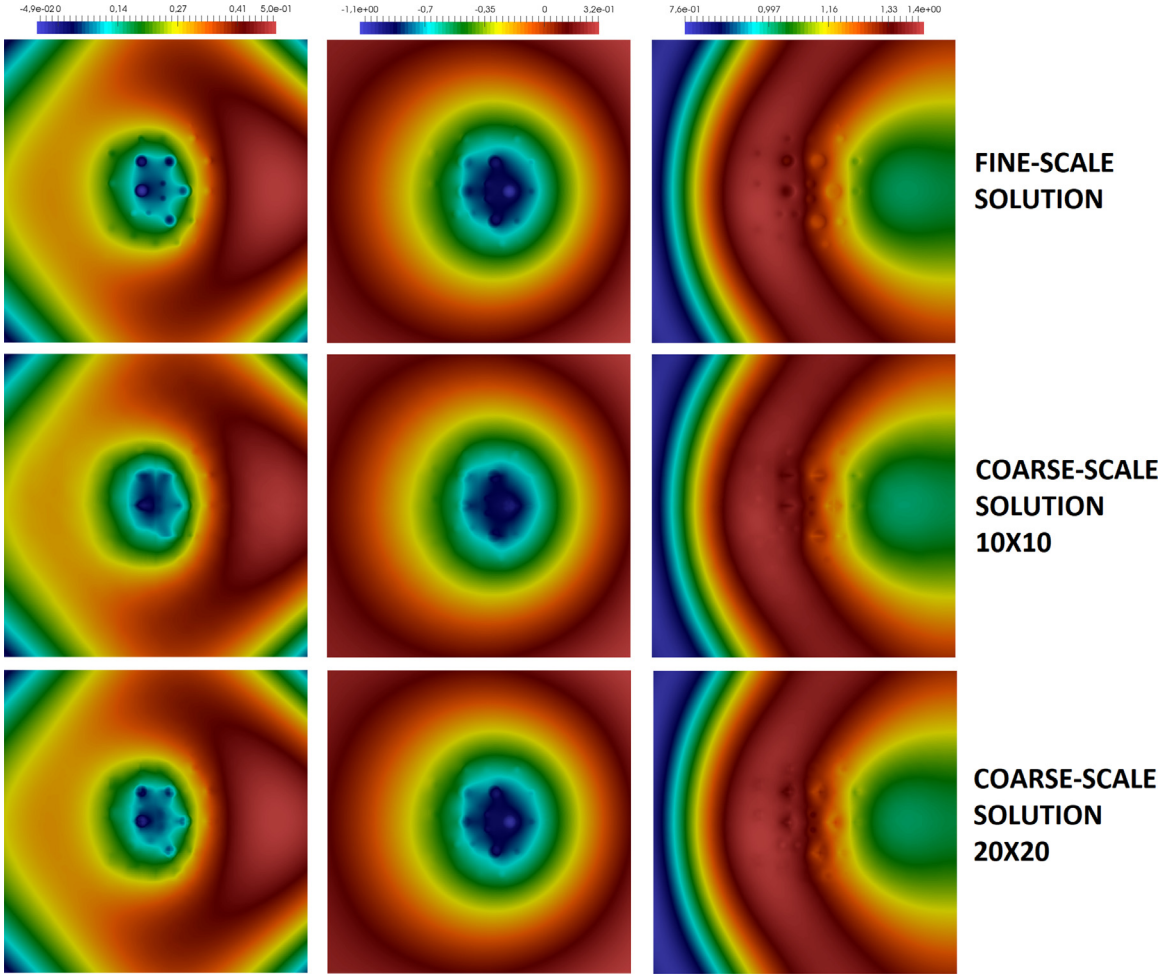


Fig. 8. Reference and multiscale solutions for $k = 1$, Re(u), Im(u) and Abs(u)(from left to right). *Type 2* multiscale basis functions with $M = 6$. First row: fine-scale, $DOF_f = 100802$. Second row: multiscale solutions on 10×10 coarse grid, $DOF_c = 726$. Third row: multiscale solutions on 20×20 coarse grid, $DOF_c = 2646$. Example 1.

Table 1
Relative errors for solution with different number of multiscale basis M . Coarse mesh with 121 local domains (10×10), $DOF_f = 100802$. Left: *Type 1* multiscale basis functions. Right: *Type 2* multiscale basis functions. Example 1.

M	DOF_c	e_{Re}	e_{Im}	e_{Abs}
$k = 0.1$				
1	121	7.433	3.993	0.0153
2	242	5.665	3.334	0.0119
4	484	3.079	1.896	0.006
6	726	2.589	1.150	0.004
8	968	2.414	1.014	0.004
12	1452	1.377	0.751	0.002
16	1936	1.242	0.616	0.002
$k = 0.5$				
1	121	7.152	3.463	0.341
2	242	5.169	2.845	0.242
4	484	3.128	1.701	0.157
6	726	2.773	1.242	0.146
8	968	2.591	1.114	0.137
12	1452	1.467	0.717	0.079
16	1936	1.327	0.612	0.071
$k = 1$				
1	121	9.885	9.707	2.917
2	242	7.552	7.382	2.236
4	484	4.962	4.372	1.407
6	726	4.466	3.795	1.260
8	968	4.134	3.507	1.164
12	1452	2.287	1.965	0.644
16	1936	2.080	1.778	0.585
$k = 2$				
1	121	40.185	39.993	16.355
2	242	29.123	28.042	10.911
4	484	20.908	21.556	8.150
6	726	18.499	19.141	7.063
8	968	17.133	17.644	6.525
12	1452	11.124	11.454	4.166
16	1936	9.873	10.147	3.670

M	DOF_c	e_{Re}	e_{Im}	e_{Abs}
$k = 0.1$				
1	121	7.433	3.993	0.015
2	242	5.662	3.334	0.011
4	484	3.080	1.900	0.006
6	726	2.587	1.150	0.004
8	968	2.412	1.014	0.004
12	1452	1.376	0.752	0.002
16	1936	1.240	0.616	0.002
$k = 0.5$				
1	121	7.151	3.463	0.341
2	242	5.089	2.832	0.237
4	484	3.106	1.747	0.155
6	726	2.717	1.235	0.143
8	968	2.529	1.103	0.134
12	1452	1.435	0.714	0.077
16	1936	1.293	0.606	0.069
$k = 1$				
1	121	6.074	6.303	1.700
2	242	4.478	4.892	1.315
4	484	3.135	2.998	0.870
6	726	2.599	2.279	0.701
8	968	2.320	1.999	0.619
12	1452	1.767	1.602	0.496
16	1936	1.576	1.411	0.439
$k = 2$				
1	121	28.242	27.848	11.045
2	242	25.430	25.030	9.469
4	484	12.500	12.742	4.373
6	726	9.144	9.187	3.080
8	968	7.606	7.592	2.574
12	1452	6.285	6.251	2.171
16	1936	5.647	5.625	1.971

3.1. Example 1

In this problem, the medium's heterogeneity is defined by obstacles (see Fig. 7(a)). We have three types of circle inclusion with different values of the coefficients

- $\alpha = 3$, $n = 0.8$ (first type),
- $\alpha = 5$, $n = 0.4$ (second type),
- $\alpha = 10$, $n = 0.2$ (third type).

Corresponded fine grid is unstructured with 100 802 vertices and 200 482 triangular elements. The coarse and fine grids are presented in Fig. 7. For multiscale approach, we consider simulations on two coarse grids: 10×10 coarse grid (see Fig. 7(b)) and 20×20 coarse grid (see Fig. 7(c)).

Fig. 8 shows the solution for a heterogeneous media with $k = 1$. The first row represents the reference solution on a fine grid, the second and third rows represent the solution on 10×10 and 20×20 coarse grid with 6 multiscale basis functions of *Type 2*. First column represent real part of solution (Re(u)), second column — imaginary part (Im(u)) and third — absolute value of solution(Abs(u)). Here DOF_c denotes the number of unknowns on the coarse grid and $DOF_c = N_v^c * M$, where M denotes the number of basis functions in each local domain and N_v^c is the number of coarse grid vertices. DOF_f is the number of unknowns on the fine grid and $DOF_f = N_v^f$, where N_v^f is the number of fine grid vertices. We observe a good solution using presented multiscale method with $e_{Abs} = 1.260\%$ on 10×10 coarse grid and $e_{Abs} = 0.517\%$ on 20×20 coarse grid.

In Table 1 and 2, we present the relative L_2 errors and computational time for different $k = 0.1, 0.5, 1, 2$ on 10×10 and 20×20 coarse grids, respectively. We investigate influence of the number of multiscale basis functions of *Type 1* and *Type 2* to the method accuracy. On the left table, we present errors for multiscale solution using *Type 1* multiscale basis functions. On the right table, we present errors for *Type 2* basis functions that take into account wavenumber k and heterogeneity of coefficient $n(x)$.

From the numerical results for 10×10 coarse grid presented in Table 1, we observe good accuracy with the same errors for the both types of multiscale basis functions for small values of $k = 0.1$ and 0.5 on 10×10 coarse grid. For

Table 2
Relative errors for solution with different number of multiscale basis M. Coarse mesh with 441 local domains (20×20), $DOF_f = 100802$. Left: *Type 1* multiscale basis functions. Right: *Type 2* multiscale basis functions. Example 1.

M	DOF_c	e_{Re}	e_{Im}	e_{Abs}
$k = 0.1$				
1	441	5.337	3.743	0.011
2	882	3.735	2.300	0.007
4	1764	2.057	0.907	0.004
6	2646	1.111	0.604	0.002
8	3528	0.950	0.511	0.001
12	5292	0.776	0.325	0.001
16	7056	0.710	0.248	0.001
$k = 0.5$				
1	441	4.171	2.723	0.197
2	882	3.297	1.847	0.159
4	1764	2.147	0.917	0.111
6	2646	1.181	0.568	0.061
8	3528	1.016	0.481	0.054
12	5292	0.839	0.343	0.044
16	7056	0.767	0.283	0.041
$k = 1$				
1	441	5.726	5.644	1.654
2	882	5.000	4.730	1.456
4	1764	3.396	2.931	0.957
6	2646	1.818	1.607	0.517
8	3528	1.544	1.361	0.436
12	5292	1.265	1.123	0.356
16	7056	1.138	1.012	0.319
$k = 2$				
1	441	25.194	25.772	9.789
2	882	21.605	22.049	8.074
4	1764	15.005	15.299	5.568
6	2646	8.396	8.591	3.050
8	3528	7.450	7.629	2.712
12	5292	6.330	6.382	2.275
16	7056	5.701	5.712	2.030

M	DOF_c	e_{Re}	e_{Im}	e_{Abs}
$k = 0.1$				
1	441	5.324	3.742	0.011
2	882	3.721	2.299	0.007
4	1764	2.042	0.907	0.004
6	2646	1.104	0.604	0.002
8	3528	0.944	0.510	0.001
12	5292	0.771	0.325	0.001
16	7056	0.705	0.248	0.001
$k = 0.5$				
1	441	3.782	2.527	0.188
2	882	2.861	1.598	0.143
4	1764	1.769	0.644	0.094
6	2646	0.946	0.433	0.051
8	3528	0.817	0.368	0.044
12	5292	0.674	0.239	0.036
16	7056	0.616	0.184	0.033
$k = 1$				
1	441	2.327	2.852	0.727
2	882	1.747	1.984	0.539
4	1764	1.183	0.965	0.307
6	2646	0.640	0.556	0.165
8	3528	0.566	0.471	0.143
12	5292	0.455	0.313	0.110
16	7056	0.412	0.241	0.098
$k = 2$				
1	441	11.990	11.530	3.339
2	882	9.134	8.609	2.482
4	1764	4.654	3.848	1.605
6	2646	2.488	2.021	0.785
8	3528	2.010	1.590	0.664
12	5292	1.367	1.034	0.430
16	7056	1.168	0.834	0.364

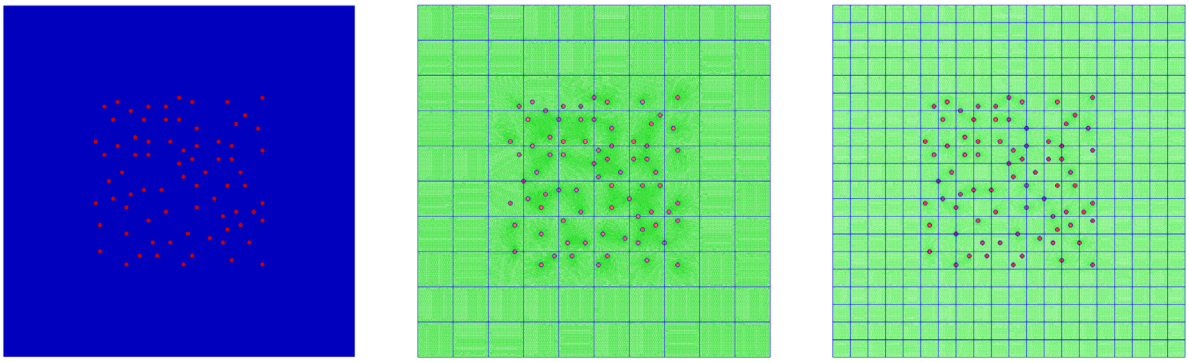
Table 3
Solution time for fine-scale (reference) solution and multiscale solution. Example 1.

M	DOF	e_{Abs}	$t_{sol,c}$
10×10			
1	121	1.700	0.004
2	242	1.315	0.008
4	484	0.870	0.049
6	726	0.701	0.154
8	968	0.619	0.344
12	1452	0.496	1.088
16	1936	0.439	2.506
Fine-scale	100802		15.341

M	DOF	e_{Abs}	$t_{sol,c}$
20×20			
1	441	0.727	0.018
2	882	0.539	0.067
4	1764	0.307	0.355
6	2646	0.165	1.231
8	3528	0.143	3.289
12	5292	0.110	11.401
16	7056	0.098	27.135
Fine-scale	100802		15.341

the cases with $k = 1$ and 2, we obtain better results using *Type 2* multiscale space that take into account k and $n(x)$. For example, we have 4.1, 3.5, 1.1% of errors for e_{Re}, e_{Im}, e_{Abs} on 10×10 coarse grid with 8 multiscale basis functions of *Type 1* for $k = 1$. For *Type 2* basis functions, we obtain better results with 2.3, 1.9, 0.6% of errors for e_{Re}, e_{Im}, e_{Abs}. For $k = 2$ on 10×10 coarse grid, we have 17.1, 17.6, 6.5% of errors for e_{Re}, e_{Im}, e_{Abs} for *Type 1* ($M = 8$) and 7.6, 7.5, 2.5% for *Type 2* multiscale space. Using 8 multiscale basis functions, we have $DOF_c = 968$ on the 10×10 coarse grid and on fine grid $DOF_f = 100802$.

Numerical results for 20×20 coarse grid are shown in Table 2. We can see that presented multiscale method works well for $k = 0.1, 0.5$ and 1 for both types of multiscale basis functions. For $k = 2$, we obtain better results for *Type 2* multiscale basis functions. For example for $M = 8$, we have 8.3, 8.5, 3.0% of errors for e_{Re}, e_{Im}, e_{Abs} for *Type 1* ($M = 8$) and 2.0, 1.5, 0.6% for *Type 2* multiscale space. Using 8 multiscale basis functions, we have $DOF_c = 3528$ on the 20×20 coarse grid and on fine grid $DOF_f = 100802$. Numerical solution using 8 multiscale basis functions on 20×20 coarse grid provide smaller error ($e_{Abs} = 2.5\%$) then on 10×10 coarse grid with same number of basis functions ($e_{Abs} = 0.6\%$). But the size of the system is larger, $DOF_c = 3528$ on the 20×20 coarse grid and $DOF_c = 968$ on the 10×10 coarse



(a) Computational domain. (b) 10×10 coarse mesh. (c) 20×20 coarse mesh.

Fig. 9. Computational domain, fine and coarse meshes. (a) Computational domain with obstacles $\Omega = \Omega_0 \cup D$. Ω_0 is the background media (blue color). D is the obstacles (red color). Fine mesh with 111216 vertices and 221310 triangle elements (green color for background media and red color for obstacles). (b) 10×10 coarse mesh with 121 vertices and 100 quadratic elements (blue color). (c) 20×20 coarse mesh 441 vertices and 400 quadratic elements (blue color). Example 2.

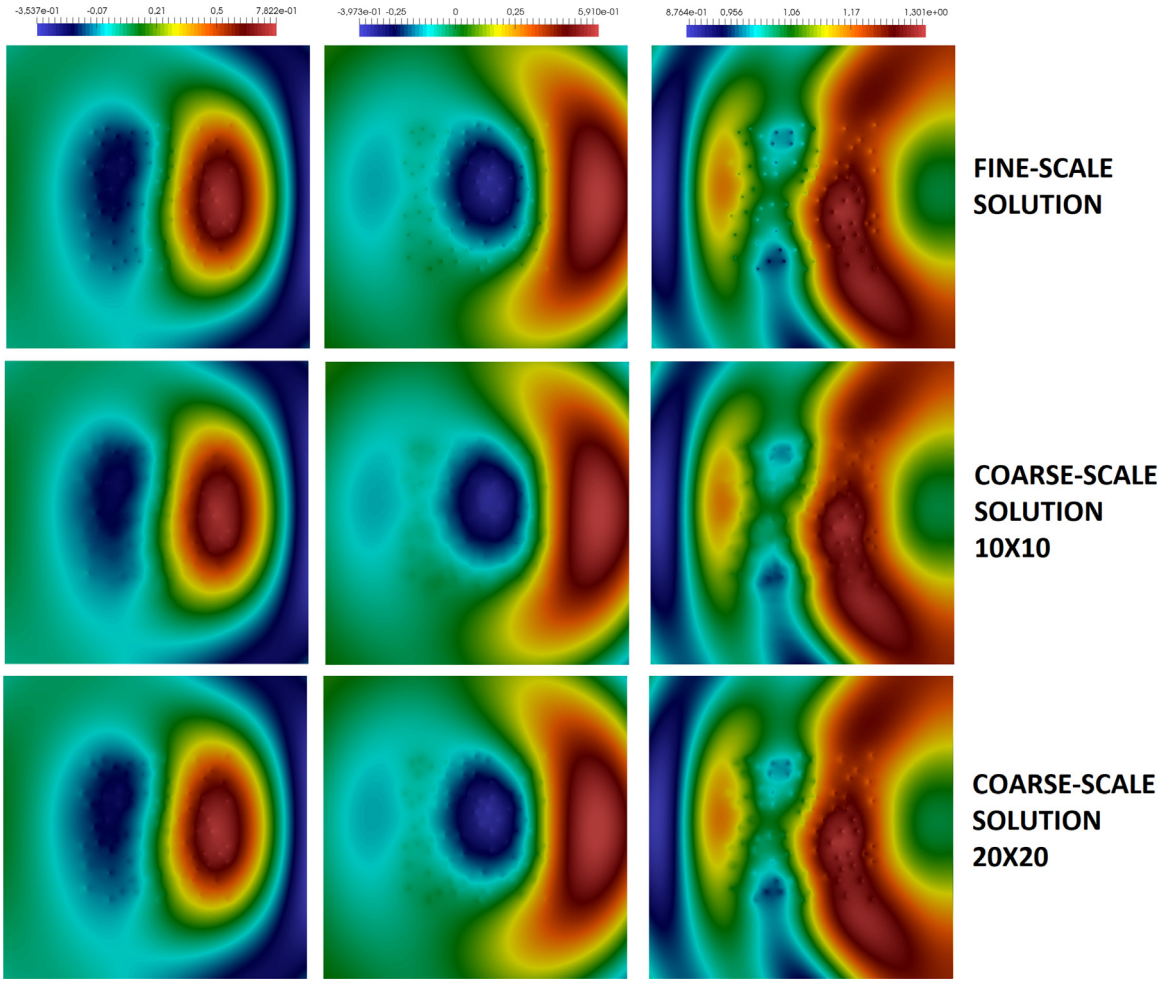


Fig. 10. Reference and multiscale solutions for $k = 2$, Re(u), Im(u) and Abs(u)(from left to right). Type 2 multiscale basis functions with $M = 16$. First row: fine-scale, $DOF_f = 111216$. Second row: multiscale solutions on 10×10 coarse grid, $DOF_c = 1936$. Third row: multiscale solutions on 20×20 coarse grid, $DOF_c = 7056$. Example 2.

Table 4
Relative errors for solution with different number of multiscale basis M. Coarse mesh with 121 local domains (10×10), $DOF_f = 111216$. Left: *Type 1* multiscale basis functions. Right: *Type 2* multiscale basis functions. Example 2.

M	DOF_c	e_{Re}	e_{Im}	e_{Abs}
$k = 0.1$				
1	121	3.5	2.578	0.002
2	242	2.865	2.335	0.002
4	484	1.853	2.075	0.001
6	726	1.455	1.884	0.001
8	968	1.323	1.789	0.001
12	1452	1.099	1.537	0.001
16	1936	1.011	1.429	0.001
$k = 0.5$				
1	121	3.499	2.254	0.062
2	242	2.778	2.018	0.049
4	484	1.635	1.613	0.024
6	726	1.328	1.473	0.019
8	968	1.241	1.405	0.017
12	1452	1.082	1.215	0.015
16	1936	0.996	1.129	0.014
$k = 1$				
1	121	4.096	3.594	0.261
2	242	3.303	2.835	0.214
4	484	1.498	1.421	0.098
6	726	1.217	1.211	0.082
8	968	1.147	1.154	0.076
12	1452	1.001	0.996	0.068
16	1936	0.922	0.923	0.062
$k = 2$				
1	121	13.521	12.273	3.117
2	242	9.785	8.871	2.249
4	484	3.232	3.29	0.718
6	726	2.478	2.587	0.541
8	968	2.364	2.461	0.509
12	1452	2.165	2.251	0.457
16	1936	2.004	2.083	0.424

M	DOF_c	e_{Re}	e_{Im}	e_{Abs}
$k = 0.1$				
1	121	3.505	2.575	0.003
2	242	2.875	2.334	0.002
4	484	1.851	2.076	0.001
6	726	1.455	1.884	0.001
8	968	1.322	1.789	0.001
12	1452	1.098	1.537	0.0005
16	1936	1.011	1.429	0.0005
$k = 0.5$				
1	121	3.191	2.073	0.061
2	242	2.541	1.871	0.048
4	484	1.582	1.595	0.024
6	726	1.296	1.462	0.019
8	968	1.208	1.393	0.017
12	1452	1.058	1.207	0.015
16	1936	0.975	1.121	0.013
$k = 1$				
1	121	2.528	2.504	0.205
2	242	1.937	1.877	0.159
4	484	1.209	1.161	0.086
6	726	1.001	1.038	0.072
8	968	0.932	0.985	0.066
12	1452	0.828	0.857	0.059
16	1936	0.761	0.792	0.054
$k = 2$				
1	121	6.802	6.024	1.486
2	242	5.173	4.219	1.121
4	484	1.966	1.803	0.423
6	726	1.438	1.374	0.294
8	968	1.284	1.237	0.255
12	1452	1.128	1.099	0.222
16	1936	1.018	1.002	0.201

grid. However, we observe that on the finer coarse grid the smaller number of basis functions can provide a good results. For example with $M = 2$, we have good results with 9.1, 8.6, 2.4% of errors for e_{Re}, e_{Im}, e_{Abs} for *Type 2* multiscale basis functions with $DOF_c = 882$.

In order to compare our method with another multiscale method, we present our previous work [24] where we consider the multiscale finite element method (MsFEM) for solving the problem of wave propagation in heterogeneous media. To construct a coarse grid approximation, local problems are also solved by two approaches to constructing local basis functions based on two types of local problems using the elliptic part of the operator and using the Helmholtz operator. When we use 1 multiscale basis function in the GMsFEM method, this is actually the MsFEM method. In the MsFEM method, the best results are shown when solving the problem on a coarse grid of 20×20 using multiscale basis functions of Type 2 (at $k = 0.1$ $e_{Re} = 5.324\%$ $e_{Im} = 3.742\%$, $k = 1$ $e_{Re} = 2.327\%$ $e_{Im} = 2.852\%$, $k = 2$ $e_{Re} = 11.99\%$ $e_{Im} = 11.53\%$). And in the GMsFEM method, where more multiscale basis functions ($M = 16$) are built, using the same parameters, that is, a coarse grid of 20×20 and Type 2 basis functions, we obtain the following errors for $k = 0.1$ $e_{Re} = 0.705\%$ $e_{Im} = 0.248\%$, $k = 1$ $e_{Re} = 0.412\%$ $e_{Im} = 0.241\%$, $k = 2$ $e_{Re} = 1.168\%$ $e_{Im} = 0.834\%$. Comparing the results, one can see that the proposed GMsFEM method takes into account inhomogeneities better and gives a more accurate approximation of the problem.

3.2. Example 2

In this section we consider computational domain with small randomly located heterogeneities ranging $[-1, -1] \times [1, 1]$ where we set $\alpha = 10$ and $n = 0.2$ (see Fig. 9(a)). The fine grid contains 111216 vertices and 221310 triangular cells. The coarse grid is uniform, which contains 121 vertices and 100 cells for 10×10 mesh and 441 vertices and 400 cells for 20×20 mesh, respectively.

In Fig. 10, we present the fine-scale and coarse-scale solution for the real part, imaginary part and absolute value with wave number $k = 2$. The first row represents the reference solution on a fine grid, the second and third rows represent the solution on 10×10 and 20×20 coarse grid with 16 multiscale basis functions of Type 2.

Table 5
Relative errors for solution with different number of multiscale basis M. Coarse mesh with 441 local domains (20×20), $DOF_f = 111216$. Left: *Type 1* multiscale basis functions. Right: *Type 2* multiscale basis functions. Example 2.

M	DOF_c	e_{Re}	e_{Im}	e_{Abs}
$k = 0.1$				
1	441	3.324	2.471	0.003
2	882	2.822	2.181	0.002
4	1764	1.732	1.656	0.001
6	2646	1.143	1.341	0.001
8	3528	0.962	1.219	0.001
12	5292	0.677	0.912	0.0003
16	7056	0.564	0.719	0.0003
$k = 0.5$				
1	441	2.424	1.803	0.039
2	882	2.048	1.609	0.033
4	1764	1.285	1.251	0.019
6	2646	0.968	1.044	0.013
8	3528	0.857	0.956	0.012
12	5292	0.666	0.722	0.009
16	7056	0.558	0.571	0.008
$k = 1$				
1	441	2.273	1.962	0.123
2	882	1.951	1.676	0.103
4	1764	1.134	1.082	0.067
6	2646	0.906	0.893	0.057
8	3528	0.81.	0.808	0.051
12	5292	0.636	0.618	0.043
16	7056	0.539	0.508	0.038
$k = 2$				
1	441	5.831	5.601	1.317
2	882	4.522	4.611	1.049
4	1764	2.261	2.421	0.499
6	2646	1.907	2.028	0.419
8	3528	1.749	1.841	0.378
12	5292	1.437	1.494	0.307
16	7056	1.25	1.303	0.269

M	DOF_c	e_{Re}	e_{Im}	e_{Abs}
$k = 0.1$				
1	441	3.319	2.471	0.003
2	882	2.816	2.181	0.002
4	1764	1.728	1.656	0.001
6	2646	1.141	1.341	0.0008
8	3528	0.959	1.219	0.0006
12	5292	0.675	0.912	0.0003
16	7056	0.563	0.719	0.0003
$k = 0.5$				
1	441	2.294	1.755	0.039
2	882	1.931	1.565	0.033
4	1764	1.207	1.228	0.019
6	2646	0.913	1.029	0.013
8	3528	0.811	0.943	0.012
12	5292	0.641	0.715	0.009
16	7056	0.535	0.564	0.008
$k = 1$				
1	441	1.652	1.9421	0.115
2	882	1.382	1.215	0.097
4	1764	0.813	0.873	0.064
6	2646	0.679	0.761	0.053
8	3528	0.621	0.695	0.048
12	5292	0.521	0.547	0.039
16	7056	0.437	0.441	0.034
$k = 2$				
1	441	3.902	3.367	0.897
2	882	3.151	2.728	0.683
4	1764	1.861	1.624	0.268
6	2646	1.725	1.519	0.239
8	3528	1.508	1.354	0.216
12	5292	1.244	1.117	0.181
16	7056	1.052	0.942	0.157

In Tables 4 and 5, we present the relative L_2 errors and computational time for different $k = 0.1, 0.5, 1, 2$ and number of the multiscale basis functions of Type 1 and Type 2 on the coarse grid 10×10 and 20×20 between multiscale and fine grid solutions. In the Table 4 (left) we present the relative L_2 errors for Type 1 multiscale basis functions. The results show that 6 multiscale basis functions are enough to achieve good results with $1.2 - 2.6\%$ of relative L_2 error for real and imaginary part of solution. The relative errors results using Type 2 multiscale basis functions are shown in Table 4 (right). It can be seen that the error improved when using Type 2 multiscale basis functions in all considered cases of the wave number. For $k = 0.1$ and 0.5 the difference between Type 1 and 2 is minimal, and for $k = 1$ and 2 Type 2 gives smaller errors. For example, for $k = 2$, the real solution Re(u) with 16 basis functions of Type 1 results in errors of $e_{Re} = 2.004\%$ ($DOF_c = 1936$), while, as for Type 2, we can obtain similar errors even with 4 basis functions of $e_{Re} = 1.966\%$ ($DOF_c = 484$), and with 16 basis functions, the error is $e_{Re} = 1.018\%$ ($DOF_c = 1936$). More accurate results are obtained in Table 5 when constructing multiscale basis functions of Type 2 on 20×20 coarse grid. For example, for $k = 0.1$, the errors are halved, where for a 20×20 grid, $e_{Re} = 0.564\%$, and for a 10×10 grid, $e_{Re} = 1.011\%$ using 16 multiscale Type 1 basis functions. For wavenumber $k = 0.1, 0.5, 1$, Type 2 does not give an improvement in error in contrast to $k = 2$ where the solution with multiscale Type 2 basis functions gives better errors than Type 1.

We obtain a good results with significant reduction of the system size for presented multiscale method with near 1% of error using *Type 2* multiscale space and sufficient number of multiscale basis functions on 20×20 coarse grid. We observe that the errors decrease as the number of multiscale basis functions increases and/or size of the coarse grid increases, as expected.

In addition, we study the computation time for multiscale solution in Tables 3 and 6 to emphasize the advantage of the method in gaining time. Computation time of the multiscale solution is much less than computation time of reference solution. We use the maximum number of basis function ($M = 16$) and we can get the solution in 2.506 s using a multiscale solver, while the time for solving on a fine grid is about 15 s in Table 3 for 10×10 coarse grid. Note that here we consider only the computational time for solving the problem on a coarse grid with preliminarily calculated basis functions, without taking into account the time of generating local domains, multiscale basis functions and constructing a system on a coarse grid. Different wave numbers and types of construction of basis functions do not affect the computational time of the problem. Larger computational gains can be obtained for 3D problems with larger fine

Table 6
Solution time for fine-scale (reference) solution and multiscale solution. Example 2.

M	*DOF*	e_{Abs}	$t_{sol,c}$	M	*DOF*	e_{Abs}	$t_{sol,c}$
10×10				20×20			
1	121	0.205	0.005	1	441	0.115	0.021
2	242	0.159	0.01	2	882	0.097	0.081
4	484	0.086	0.058	4	1764	0.064	0.496
6	726	0.072	0.167	6	2646	0.053	1.558
8	968	0.066	0.359	8	3528	0.048	3.622
12	1452	0.059	1.201	12	5292	0.039	11.983
16	1936	0.054	2.803	16	7056	0.034	28.027
Fine-scale	100802		18.614	Fine-scale	100802		18.614

mesh system size. Moreover, multiscale basis functions can be computed in parallel since all basis functions are computed independently of each other.

4. Conclusion

We considered the scattering problem in heterogeneous media with obstacles. For numerical solution, we generate fine grid that resolve obstacles on the grid level and construct discrete problem using a finite element method. In order to reduce size of the fine grid approximation, we present a multiscale method for coarse grid approximation. We utilize a Generalized Multiscale Finite Element Method (GMsFEM) for solution of scattering problem. In GMsFEM, we constructed multiscale basis functions based on the solution of the local spectral problems on the snapshot space. We present two approaches for construction of the local basis functions based on the two types of local problems using elliptic part of operator and using the Helmholtz operator. Numerical investigation is presented for model problems with different wavenumber on two coarse grids (10×10 and 20×20). The method accuracy is presented for the different number of multiscale basis functions for two types of multiscale basis functions. Numerical results showed that the presented method provided accurate solution of the problem with significant reduction of the system size.

Data availability

Data will be made available on request.

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